The covar and orig Data-Generating Processes: A Component-Wise Assessment, with Application to the $c_{bm} \leq 0.35$ Regime

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Purpose. This paper audits the stated differences between the package's covariance-moderation process (`covar`) and the vendored copy of the Hendrickson et al. [1] reference implementation used as paper 01's comparison arm, assesses each difference on its merits, and asks whether restricting c_{bm} to 0.35 or below would remove the reason to prefer `covar`.

Review note (2026-10-07): what `orig` means here. In this paper `orig` denotes the *vendored 2024 copy* of the upstream repository (HEAD `06dac83`, 2024-11-27), not the published code. Project convention has since fixed `orig` to mean commit `58b32a9` (2020-11-19), whose data-generating code is identical to the 2020 initial commit: compound symmetry, a constant cross-factor block, step coupling on a nonzero carryover-adjusted response mean at every occasion including the first, no `scalefactor`, and `lmer` with a binary drug indicator (inspected, copy at `analysis/data/quick-sim/hendrickson-58b32a9/`; see `docs/32` and `docs/36`). Read with that convention, four statements below change. (a) Paper 01's description of the step coupling was a correct description of `orig`; it was the comparison arm that ran different code. (b) Differences 4 and 5 (the dropped first occasion and the `scalefactor`) are properties of the 2024 revision, not of `orig`. (c) At the published $\rho = 0.8$ the published construct's ceiling is 0.256 (Hybrid, CO) to 0.262 (OL+BDC) without carryover, and 0.893 at any positive half-life in Hybrid and OL+BDC (`docs/36`, Section 4.2; recomputed in this review); the ceilings below are at $\rho = 0.7$ and for the 2024 coupling. Under the separable AR(1) form of `docs/35` the ceiling is 0.440 at $\rho = 0.8$, and 0.474 (Hybrid 0.480) at $\rho = 0.7$. (d) The Section 5 finding that AR(1) alone beats the full `covar` set is an artifact of the dropped first occasion (Section 5.3, note). Paper 01 has also been restructured; its Appendix B.5 to B.7 cited below refer to the September 2026 draft.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here and are not part of the canonical set. Code identifiers appear in backticks.

Table 1: Notation index

Symbol	Meaning	Status
c_{bm}	covariance-moderation correlation: on-drug correlation of the biomarker with BR (code `c.bm`)	canonical
$\tilde{r}$	biomarker coupling vector: correlations of the biomarker with BR at each occasion (called "the interaction vector b" in earlier versions; renamed because b is the canonical standardized biomarker)	local
$\mathbf{1}_n$	vector of n ones	local
n	number of measurement occasions	local
w_t, w_s	cumulative calendar week of occasions t and s	local
ρ	within-factor correlation, CS or AR(1) (code `c.tv`, `c.pb`, `c.br`)	canonical
$t_{1/2}$	carryover half-life, weeks (code `carryover_t1half`)	canonical
t_{sd}	time since discontinuation (code `tsd`)	canonical
λ	carryover decay rate, $\ln 2 / t_{1/2}$	canonical
μ_{BR}	mean of the drug-response component BR (code `brmeans`)	local
BR, PB, TV	the three response components	canonical
D_{bc}	exposure-decayed drug indicator (code column `Dbc`)	canonical
Σ	joint covariance matrix of biomarker, baseline and response components	local
M	correlation block of the three response components	local
v	coupling pattern, $\tilde{r} = c_{bm} v$	local
$c_{\times}$	cross-factor correlation at distinct occasions (code `c.cfct`)	local

Glossary. ***`orig`*** (this paper only): the vendored 2024 copy of the Hendrickson et al. code; see the review note. ***`covar`***: the package's covariance-moderation process (`dgp_architecture = 'mvn'`, formerly Architecture B). *Compound symmetry (CS)*: one constant correlation for every pair of occasions. *PD*: positive definite. *Ceiling*: the largest c_{bm} at which Σ is PD. *Silent repair*: `corpcor::make.positive.definite()` applied without warning when Σ fails the PD check. ***`scalefactor`***: multiplier on the carryover exponent added in the 2024 revision. *Step* versus *graded* coupling: c_{bm} wherever any residual drug effect remains, versus a coupling that decays off drug.

1. Executive summary

Paper 01 (`analysis/report/01-dgp-mean-moderation-vs-mvn/`) compares three data-generating processes, of which two are covariance-channel processes: `covar` (the package's Architecture B) and `orig` (a vendored copy of the Hendrickson et al. reference implementation). Appendix B.6 of that paper states that `orig` differs from `covar` on three rows simultaneously. This white paper audits that claim against the code that runs, assesses each difference on its merits, and answers a specific practical question: if the biomarker-response correlation c_{bm} is restricted to 0.35 or below, is there any remaining reason to prefer `covar` over `orig`?

There are five differences, not three, and the appendix describes the wrong version of the most consequential one. The three documented rows are the within-factor correlation form, the cross-factor off-diagonal, and the coupling vector $\tilde{r}$. Two further differences are undocumented: `orig` never assigns the biomarker-response correlation at the first measurement occasion, and it decays the response mean at twice the nominal rate through a `scalefactor` parameter that `covar` lacks. The appendix's description of $\tilde{r}$ corresponds to code that is commented out in the vendored file; the active code behaves in the opposite direction. *(With `orig` defined as `58b32a9`, as the project now does, the appendix's description is correct and the two undocumented differences belong to the 2024 revision only; see the review note.)*

The active $\tilde{r}$ code is the better of the two, contrary to first appearance. The commented-out published version raises the off-drug biomarker-response correlation to full strength as carryover grows, inverting the effect residual exposure should have. The 2024 revision repairs this. Its defects are the `scalefactor` default and the dropped occasion, not the change of gate itself (Section 3.3.1).

Only one of the four covariance-matrix changes is required to fix positive definiteness. A crossed decomposition over all 16 combinations of those four (Section 5) shows that switching compound symmetry to AR(1) accounts for the entire improvement, raising the worst-cell c_{bm} ceiling from 0.34 to 0.48. Applied alone it beats the full four-change `covar` set, which reaches only 0.45. The cross-factor change lowers the ceiling to 0.16 when applied without AR(1), and neither the $\tilde{r}$ change nor the occasion-1 change moves the worst cell when AR(1) is off. *(Corrected 2026-10-07: the advantage of AR(1) alone over the full set comes from leaving the first, on-drug occasion uncoupled; with it coupled, AR(1) alone reaches 0.438, below `covar`'s 0.452. The occasion-1 change does move the ceiling once AR(1) is on. Section 5.3 and 5.5 notes.)*

The answer to the practical question is yes, on five independent grounds. Restricting c_{bm} to 0.35 does not rescue `orig`. The positive-definite (PD) ceiling under `orig`, on a grid of step 0.05, is 0.30 in five of nine design-by-half-life cells and 0.35 in the remaining four, so a study run at 0.35 is non-PD in a substantial part of its own grid. Where it is PD at 0.35 the margin is

thin enough to be fragile to any other parameter change. PD failure is repaired silently, so the realized correlation stops matching the nominal one without any warning. The dropped first occasion is a defect at every c_{bm}, including zero. And the `scalefactor` default halves the effective carryover half-life, so the two processes are not comparable at a matched nominal half-life regardless of correlation strength.

The vendored analysis model was also replaced, and carries a separate correctness bug. The `orig` arm does not run the published analysis either: the binary `Db` regressor became the continuous `Dbc` in 2024. In that code every never-treated occasion, baseline included, is coded as fully on drug whenever `carryover_t1half > 0`, so the crossover design's never-discontinuing path contributes no on-drug versus off-drug contrast at all. The package's own `R/lme_analysis.R` already guards this case, so no result computed through the package pipeline is affected (Section 3.6).

The remedy is a corrected `orig`, not a choice between the two. Three of the five 2024 DGP changes are sound and should be kept; the `scalefactor` default and the dropped occasion should be fixed; and the single AR(1) change of Section 5 resolves positive definiteness outright (Section 4.5).

2. Scope, method, and evidence status

Findings below are labeled `verified` (code was executed and output observed), `inspected` (read in source and confirmed by reading), or `inferred`. Nothing in this document is asserted from memory of the manuscripts.

Sources examined:

- `analysis/scripts/quick-sim/hendrickson-original-comparison/` (`vendored-hendrickson-generateData.R`, `01-hendrickson-orig-driver.R`, `02-pd-sweep.R`, `README.md`)
- `R/generateData.R` (`buildSigma`, the `covar` construction) and `R/lme_analysis.R` (the package's own analysis model)
- `implementations/original/R/generateData.R`
- `analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`, Appendix sections B.4 through B.7
- The upstream repository `github.com/rchendrickson/pmsimstats`, cloned in full (21 commits, 2020-02-22 to 2024-11-27)

Computations run for this document:

- Reconstruction of the $\tilde{r}$ vector under both the active and the commented-out code paths, Hybrid path A, at $t_{1/2} \in \{0.5, 1.0\}$.

- A PD sweep over $c_{bm} \in [0.10, 0.60]$ by 0.05, crossed with $t_{1/2} \in \{0, 0.5, 1.0\}$ and the three trial designs, recording the minimum eigenvalue of Σ under each process, minimized across the design's randomization paths.
- Quantification of the distortion introduced by PD repair.
- A crossed decomposition over all 16 combinations of the four covariance-matrix changes (Section 5).
- Comparison of the carryover mean adjustment at `scalefactor` 1 and 2 against a correct exponential decay from the last on-drug value (Section 3.5).
- Diffs of the vendored files against the upstream initial commit (`42ac030`), the last pre-acceptance commit (`3035581`), and HEAD (`06dac83`).
- Evaluation of the `Dbc` exposure regressor across all occasions of the CO and Hybrid designs, under both the vendored and the package analysis code, at $t_{1/2} \in \{0, 1.0\}$ (Section 3.6).

Provenance correction. The directory README states the vendored files come from `3035581` (dated there as 2026-06-28; the commit is actually 2020-06-28 04:00 UTC, 2020-06-27 Pacific). They do not. Diffing establishes the base as HEAD, `06dac83`, 2024-11-27, from which the vendored copy differs by exactly the one documented `data.table` line. The README's claim that no other lines were changed is therefore true of the file that was vendored and false of the file it names. The `RON THOMAS` annotations are upstream, introduced by RC Hendrickson in commit `8609f12` (2024-05-06, 'Ron Thomas' edits, currated into working code'), not local edits.

All computations used the package's own `extracted_rp` and `extracted_bp` parameter sets, $\rho = 0.7$ (`c.tv`, `c.pb`, `c.br`), `c.cf1t = 0.2`, `c.cfct = 0.1`, `N = 35`, `scalefactor = 2`. The published results used $\rho = 0.8$ (review note).

3. The five DGP differences, and the analysis model

3.1 Difference 1: within-factor correlation form

Status: verified.

`orig` assigns a single constant to every within-factor occasion pair (`vendored-hendrickson-generateData.R:117-126`):

```
ac <- modelparam[[paste("c", c, sep = ".")]]
for (p in 1:(nP-1)) for (p2 in (1+p):nP) {
  correlations[n1, n2] <- ac
  correlations[n2, n1] <- ac
}
```

This is compound symmetry. `covar` uses $\rho^{|w_t - w_s|}$, an AR(1) form on the cumulative week scale.

Arguments for compound symmetry. It is the form in the published reference, so it is the correct target if the goal is reproducing Hendrickson et al. It has one parameter and is exchangeable, which makes it invariant to occasion ordering and therefore robust to irregular visit spacing. It is the standard random-intercept implication, so it is internally consistent with an analysis model that fits only a patient random intercept.

Arguments against. It is clinically implausible for repeated symptom measurement, where adjacent occasions should correlate more strongly than distant ones. More seriously for present purposes, it constrains the feasible parameter space. A constant ρ across n occasions requires $\rho > -1/(n-1)$ on its own, and once the biomarker-response block is added the joint constraint on c_{bm} tightens rapidly with n. Appendix B.7 identifies this as the mechanism behind the narrower PD range, and Section 4 below quantifies it. (Precisely: the response block's smallest eigenvalue does not depend on n, but the ceiling on c_{bm} falls roughly as $n^{-1/2}$ at a fixed on-drug share, `docs/34`, Section 5.1. At $\rho = 0.3$ compound symmetry admits the higher ceiling, `docs/34`, Section 4.4.)

Assessment. This difference is the dominant driver of the PD behavior and therefore of the answer to the paper's practical question. It is a legitimate modeling choice in isolation, but it is not a neutral one: it changes what parameter values can be studied at all.

3.2 Difference 2: cross-factor off-diagonal

Status: verified.

`orig` assigns `modelparam$c.cfct` as a constant to every cross-factor pair at distinct occasions (line 140). `covar` assigns `c.cfct * rho^tg`, that is $c_{\times}\rho^{|w_t - w_s|}$, decaying in the lag `tg`.

Arguments for the constant form. Again, fidelity to the reference. It also imposes no assumption about how quickly the coupling between distinct latent factors decays, which is a quantity for which little empirical guidance exists.

Arguments against. It is internally inconsistent with any decaying within-factor structure, since it asserts that a factor's correlation with a different factor eight occasions away is as strong as with itself at the adjacent occasion. It contributes to the PD constraint in the same direction as Difference 1, since constant off-diagonal blocks raise the matrix's dependence on its smallest eigenvalue.

Assessment. Secondary to Difference 1 in magnitude, and in the same direction. It is not separately identifiable from Difference 1 in any comparison run to date, since the two always vary together.

Correction (2026-10-07). The crossed decomposition of Section 5, written after this paragraph, does separate the two, and contradicts it. Under compound symmetry the decaying cross-factor form lowers the ceiling (0.34 to 0.16), so the constant form of `orig` is the better of the two there; under AR(1) the decaying

form raises it slightly (about 0.02 to 0.03; `docs/34`, Table 4.1). The constant form is therefore not in the same direction as Difference 1 in general.

3.3 Difference 3: the coupling vector $\tilde{r}$

Status: verified, and the paper's description is incorrect *(for the vendored 2024 copy; it is correct for `58b32a9`, review note).*

Appendix B.6 states that `orig` gates the interaction on whether the response mean is nonzero, giving a two-valued step of $\{0, c_{bm}\}$, and that at $t_{1/2} = 1.0$ in the Hybrid design the vector becomes $c_{bm}\mathbf{1}_n$, carrying no on-drug versus off-drug contrast at all. The worked example at B.5 prints $\tilde{r}^{\text{orig}} = (0.4500, 0.4500, 0.4500, 0.4500)$.

That describes the following code, which is present in the vendored file but **commented out**, under the header `## Following commented out in Ron Thomas version:`

```
#if(means[which(n1==labels)]!=0){
#  correlations[n1,'bm']<-modelparam$c.bm
#  correlations['bm',n1]<-modelparam$c.bm
#}
```

The code that actually executes, under the header `## RON THOMAS VERSION:`, is line 153:

```
correlations["bm", n1] <- correlations[n1, "bm"] <-
  ifelse(brtest[p],
         ifelse(brmeans[p] == 0, 0, (mm1 / mm0) * modelparam$c.bm),
         modelparam$c.bm)
```

where `mm1` and `mm0` are the carryover-adjusted response means at occasions `p` and `p - 1`, and `brtest` is computed before the carryover adjustment, so it is TRUE exactly at off-drug occasions.

Re-running both code paths over the Hybrid path A design gives:

Table 2: Adjusted μ_{BR} and coupling $\tilde{r}$ by occasion, active versus commented-out code, Hybrid path A

occasion	on drug	adjusted μ_{BR}	$\tilde{r}$ active	$\tilde{r}$ commented-out
OL1	yes	4.281	NA	0.45
OL2	yes	9.223	0.4500	0.45

occasion	on drug	adjusted μ_{BR}	$\tilde{r}$ active	$\tilde{r}$ commented-out
BD1	yes	9.793	0.4500	0.45
BD2	yes	10.187	0.4500	0.45
BD3	no	2.547	0.1125	0.45
BD4	no	0.159	0.0281	0.45
COd	yes	4.281	0.4500	0.45
COp	no	0.017	0.0018	0.45

($t_{1/2} = 1.0$, `scalefactor = 2`, $c_{bm} = 0.45$.)

The commented-out column reproduces the appendix exactly. The active column does the opposite: off-drug entries decay toward zero rather than rising to c_{bm}, and the on-drug versus off-drug contrast is preserved rather than lost.

The active decay belongs to the same exponential family as `covar`'s, but runs at twice the rate. The ratio `mm1 / mm0` inherits `(1/2)^(scalefactor * tsd / carryover_t1half)` from the carryover adjustment at line 96, and with `scalefactor = 2` this is an effective half-life of $t_{1/2}/2$. Because the ratio is taken against the previous, already adjusted mean, the coupling itself is exactly $c_{bm}(1/2)^{2t_{sd}/t_{1/2}}$ with cumulative t_{sd}; it does not compound across successive off-drug occasions, although the mean does (Section 3.5). (Corrected 2026-10-07; an earlier version said the coupling compounded. Derived, and consistent with the table: 0.1125 and 0.0281 are $0.45 \cdot 2^{-2}$ and $0.45 \cdot 2^{-4}$.) For comparison, at $t_{1/2} = 1.0$ and one week off drug, `covar` gives $0.45\, e^{-\ln 2} = 0.225$ while `orig` gives 0.1125.

3.3.1 Which of the two is correct **Status: verified.** The natural reading of the paragraphs above is that the commented-out code is the correct target and the active code an unwarranted deviation from it. On assessment that reading is wrong, and it should be stated plainly before the arguments are weighed.

The published gate tests `means[...] != 0`, where `means` is the **carryover-adjusted** response mean. The adjustment runs before the correlation loop, so at any off-drug occasion carrying residual effect, the adjusted mean is nonzero, the gate opens, and the occasion receives the **full** c_{bm}. The consequence is that increasing carryover progressively converts off-drug occasions into full-strength on-drug occasions in the covariance channel, until at $t_{1/2} = 1.0$ in the Hybrid design the vector is $c_{bm}\mathbf{1}_n$ (as at any positive half-life, `docs/37`) and no on-drug versus off-drug contrast survives at all.

That behavior is backwards. Carryover should attenuate the biomarker-response coupling at off-drug occasions, not raise it to on-drug strength. The published DGP therefore has the property that more carryover means a *stronger*, not

weaker, off-drug interaction signal, which is not a defensible representation of residual drug exposure.

The active code gates on `brtest`, computed before the carryover adjustment (Difference 5, Section 3.5), so it identifies genuinely off-drug occasions, and then grades the correlation downward rather than holding it at full strength. **The direction is correct and the published version was wrong.** Whatever its implementation defects, the 2024 revision repairs a real error rather than introducing one.

3.3.2 Assessment **Arguments for the active form.** It fixes the inversion described above, which is the substantive point. It is graded rather than a step, which is the more plausible representation. And it ties the correlation channel to the same mean trajectory the carryover adjustment already modifies, which is internally consistent.

Arguments against. Its decay rate is set by `scalefactor`, whose default of 2 is itself an error (Section 3.5). It drops occasion 1 (Section 3.4). And it is undocumented: nothing in Paper 01, the vendored file's header, or the directory README describes it, so a reader of the manuscript cannot know which behavior the `orig` arm has.

Assessment. The serious finding here is documentary rather than methodological. Paper 01's Appendix B.5 and B.6 describe the 2020 code, which the `orig` arm does not run, and the appendix's causal story for `orig`, that carryover opens its gate and destroys the identifying contrast, is an accurate account of a defect that the running code has already fixed. Any interpretation in Paper 01 resting on that mechanism requires revision. But the fix should not be to restore the 2020 behavior; it should be to describe the 2024 behavior accurately and correct its two remaining implementation flaws.

3.4 Difference 4: the first occasion is never assigned

Status: verified.

The assignment loop at line 147 is guarded by `if (p > 1)`, and `correlations` is initialized as `diag(length(labels))` at line 106. The biomarker-response correlation at occasion 1 is therefore left at exactly zero under `orig`, regardless of c_{bm} and regardless of whether occasion 1 is on drug. In the table above this appears as the `NA` in the OL1 row. `covar` assigns all eight occasions.

Arguments for. None identified. This does not appear to be a modeling choice.

Arguments against. It removes one of eight occasions from the interaction channel, and it does so at an occasion that is on drug in every path of every design examined, so the loss falls on the most informative part of the series.

It attenuates `orig`'s realized interaction relative to its nominal c_{bm} at every parameter setting. It also interacts with Difference 3: the `p > 1` guard exists because line 153 needs `p - 1` to form the ratio, so the defect is a direct consequence of the undocumented modification, not an independent bug.

Assessment. A defect rather than a difference in modeling philosophy. It is c_{bm}-independent and therefore is not addressed by restricting the correlation range.

3.5 Difference 5: the scalefactor on the carryover mean

Status: verified.

Paper 01's Appendix B.6 records the carryover adjustment to the response mean as an 'identical line in all three' processes. It is not. `orig` carries a `scalefactor` multiplier that `covar` does not:

```
## orig
brmeans[p] <- brmeans[p] +
  brmeans[p-1]*(1/2)^(scalefactor * d$tsd[p]/carryover_t1half)

## covar
brmeans[p] <- brmeans[p] +
  brmeans[p-1]*(1/2)^(d$tsd[p]/carryover_t1half)
```

`scalefactor` defaults to 2, so `orig` decays the response mean at twice the nominal rate. This is a difference in the **mean** channel, distinct from the covariance-channel differences above, and it affects any power comparison between the two processes even though it does not enter Σ and so does not appear in the Section 5 decomposition.

The default is an error, not merely a choice. Comparing the adjustment against a correct exponential decay from the last on-drug value, at $t_{1/2} = 1.0$ in the Hybrid design:

Table 3: Off-drug response-mean adjustment under scalefactor 1 and 2 versus correct decay, Hybrid design

occasion	t_{sd}	sf = 1	sf = 2	correct
BD3	1	5.094	2.547	5.094
BD4	2	1.273	0.159	2.547
COp	4	0.268	0.017	0.268

As a ratio to the correct value: at `sf = 1` the first off-drug occasion after each on-drug run is exact (1.000x at BD3 and COp) and the second over-decays

(0.500x at BD4). At `sf = 2` even the first occasion is wrong (0.500x at BD3), and COp is understated by a factor of 16 (0.0625x). *(Clarified 2026-10-08. The 0.500x holds only where $t_{sd} = t_{1/2}$. In general the ratio at the first off-drug occasion of a run is $2^{-(s_f-1)t_{sd}/t_{1/2}}$: 0.500x at BD3 ($t_{sd} = 1$) and 0.0625x at COp ($t_{sd} = 4$), which is also the first off-drug occasion of its run. See `analysis/report/compendium-mathematics.tex`, Section "The 2024 revision of the published code: a scale factor on the carryover exponent".) (Corrected 2026-10-07. An earlier version gave the correct COp value as 0.637, which decays from the BD2 mean of 10.187; but COp follows the on-drug occasion COd, whose mean of 4.281 is the last on-drug value, and $4.281 \times 2^{-4} = 0.268$. The earlier ratios 0.420x and "a factor of 38" followed from that error. Derived from the table's own inputs.)* A dose one week old at a one-week half-life should retain half its effect; the shipped default gives it a quarter.

The parameter is also undocumented in the strict sense: its Roxygen entry reads `@param scalefactor TODO update when understand what this does?`, so it was shipped with a behavior-changing default while recorded as not understood.

The residual over-decay at `sf = 1` is a separate, pre-existing bug inherited from the published code. The recursion multiplies the previous **already-adjusted** value by $(1/2)^{t_{sd}/t_{1/2}}$ with cumulative t_{sd}, double-counting elapsed time. The correct form uses either the interval rather than cumulative t_{sd}, or cumulative t_{sd} applied to the last on-drug mean. *(Note 2026-10-07: `covar` and the mean-moderation process have since been corrected to the anchored form, `R/generateData.R` lines 277 to 288, inspected; paper 01 applies the same correction to `58b32a9`, its Appendix B.3.)*

Arguments for. None identified for the default of 2. The parameter itself is a harmless generalization if defaulted to 1.

Arguments against. It breaks a case the published code got right, it compounds an existing over-decay rather than fixing it, and it silently changes the DGP for every caller that does not override it.

Assessment. This is the change that should be removed. Setting `scalefactor = 1` restores correct behavior at the first off-drug occasion at no cost; fixing the recursion would correct the rest.

3.6 The analysis model: lme_analysis

Status: verified.

The five differences above concern `generateData.R`. The comparison arm vendors a second file, `vendored-hendrickson-lme_analysis.R`, which supplies the analysis model. It is byte-identical to upstream HEAD, so the README's statement that it is unpatched is exactly true. Five commits changed it between `3035581` and HEAD, altering 110 lines. Three findings follow.

3.6.1 The analysis model itself was replaced The published analysis and the running analysis test different coefficients:

Table 4: Analysis model of the 2020 publication versus the 2024 running code

	2020 (`3035581`)	2024 (HEAD, what runs)
exposure regressor	`Db`, binary `tod > 0`	`Dbc`, continuous decay
formula term	`bm*Db`	`bm*Dbc`
coefficient extracted	`bm:DbTRUE`	`bm:Dbc`

The change came in `325314f` (2024-05-10, "modified Db to include carryover; no effect"), four days after the DGP changes; `f70f86d` (2024-05-06) had refactored the formula construction and added the `simplecarryover` option, and `78c8635` (2024-11-27) corrected the extracted name from `bm:DbcTRUE` to `bm:Dbc`. *(Corrected 2026-10-07, verified against the upstream commit diffs with `gh api`; an earlier version attributed the change to `f70f86d`.)* The `orig` arm therefore does not run the published analysis either. Its exposure regressor is structurally the same object this program calls the exposure-weighted specification, E2 (`analysis/report/NOTATION.md`, Part 3; the earlier label `G3` is retired).

This **corroborates** the retraction in Paper 02, Section 3.3. That section argues the reference analysis is effectively Unadjusted because the drivers leave the analysis-side half-life at its default of zero, at which the decayed predictor collapses onto the binary indicator. Verified directly: at `carryover_t1half = 0` the off-drug branch evaluates `(1/2)^(tsd/0) = 0` and the on-drug branch 1, which is exactly `Db`. The reasoning in that section holds.

3.6.2 Never-treated occasions are coded as fully on drug `Dbc` is assigned to every `Db == FALSE` row as `(1/2)^(carryover_scalefactor * tsd / carryover_t1half)`. That expression is meaningful only after discontinuation. For occasions before a patient's first dose, `buildtrialdesign` zeroes `tsd` (multiplying by `everondrug`), so the expression returns `(1/2)^0 = 1`, the **maximum** exposure value.

CO path B, the never-discontinuing arm, at $t_{1/2} = 1.0$:

Table 5: Drug indicator D_{bc} (`Dbc`) by occasion for CO path B at $t_{1/2} = 1.0$

occasion	on drug	`tsd`	`Dbc`
BL	FALSE	0	**1.0**
COa1 - COa4	FALSE	0	**1.0**
COb1 - COb4	TRUE	0	1.0

All nine rows are coded `Dbc = 1`. Four pre-treatment occasions and baseline are indistinguishable from fully on drug, the path contributes no on-drug versus off-drug contrast at all, and untreated baseline data enter the model as treated. Under Hybrid the damage is one row per patient rather than five, but it is present in every design whenever `carryover_t1half > 0`.

At `carryover_t1half = 0` the same rows evaluate `(1/2)^(0/0)` and return `NaN`, so baseline is silently dropped by `na.omit` instead of miscoded. Different symptom, same root cause. The current driver passes no `carryover_t1half` and so hits the `NaN` branch, which is why this has not surfaced as a visible failure.

This is a genuine correctness bug, upstream, and distinct from anything in `generateData.R`.

3.6.3 The package's own lme_analysis is not affected Status: verified. `R/lme_analysis.R` in this package already guards the case, and its comment names the failure mode precisely:

```
# Guard against 0/0 = NaN when carryover_t1half == 0 or
# tsd == 0 for never-on-drug subjects; both collapse to Dbc=0.
if (op$carryover_t1half == 0) {
  data.m2[Db==FALSE, Dbc:=0]
} else {
  data.m2[Db==FALSE & tsd<=0, Dbc:=0]
  data.m2[Db==FALSE & tsd>0,
          Dbc:=((1/2)^(op$carryover_scalefactor*tsd/op$carryover_t1half))]
}
```

Comparing the two on CO path B:

Table 6: `Dbc` from the package and vendored code on CO path B by half-life

$t_{1/2}$	package	vendored
0	0 0 0 0 0 1 1 1 1	NaN NaN NaN NaN NaN 1 1 1 1
1.0	0 0 0 0 0 1 1 1 1	1 1 1 1 1 1 1 1 1

Finding 3.6.2 is therefore confined to the vendored comparison arm. No result in Paper 01 or Paper 02 that uses the package's own analysis pipeline is affected by it.

3.6.4 Minor `error(...)` is called in the guard against combining `simplecarryover` with a half-life-based carryover. No such function exists in

base R (verified), so the guard halts with `could not find function "error"` rather than its intended message. It was evidently never exercised.

Assessed and found unimportant: the `union(timeptnames, "BL")` change, which prevents `BL` being selected twice with no behavioral effect otherwise; and the refactor of the model formula from nested `if` blocks to string concatenation with `eval(parse(...))`, which is self-described in the source as 'poor programming form!' but is equivalent. `baf4127`, titled 'first commit. edit to lme_analysis', does not touch the file at all (it edits a vignette), and `1cbdb39` changes one line, repairing `op*carryover_scalefactor` to `op$carryover_scalefactor` in the `Dbc` assignment. *(Corrected 2026-10-07, verified with `gh api`: an earlier version said `325314f` changed only whitespace in this file; it is the commit that introduced `Dbc` and the `error(...)` guard, Section 3.6.1.)*

4. The $c_{bm} \leq 0.35$ question

4.1 Why the question arises

Compound symmetry constrains the feasible correlation range more tightly than AR(1), as Appendix B.7 notes, at the production $\rho = 0.7$; at $\rho = 0.3$ the order reverses (`docs/34`, Section 4.4). If the binding objection to `orig` were only that it cannot represent large c_{bm}, then restricting attention to $c_{bm} \leq 0.35$ would neutralize the objection, and the choice between processes would come down to fidelity to the published reference, which would favor `orig`.

4.2 Positive-definiteness sweep

Status: verified. Minimum eigenvalue of Σ, minimized across each design's randomization paths, at the parameter settings in Section 2.

Largest c_{bm} on the tested grid at which Σ remains PD:

Table 7: Largest positive-definite c_{bm} on the tested grid by design and half-life, orig and covar

design	$t_{1/2}$	orig	covar
CO	0.0	0.30	0.60
CO	0.5	0.30	0.60
CO	1.0	0.30	0.60
Hybrid	0.0	0.30	0.45
Hybrid	0.5	0.35	0.50
Hybrid	1.0	0.35	0.50
OL+BDC	0.0	0.30	0.45
OL+BDC	0.5	0.35	0.45

design	$t_{1/2}$	orig	covar
OL+BDC	1.0	0.35	0.50

Minimum eigenvalue at exactly $c_{bm} = 0.35$:

Table 8: Minimum eigenvalue of Σ at $c_{bm} = 0.35$ by design and half-life, orig and covar

design	$t_{1/2}$	orig	covar	orig PD
CO	0.0	-0.3069	8.516	no
CO	0.5	-0.3069	8.516	no
CO	1.0	-0.3069	8.516	no
Hybrid	0.0	-0.3140	2.158	no
Hybrid	0.5	0.0831	2.297	yes
Hybrid	1.0	1.2913	2.340	yes
OL+BDC	0.0	-0.3217	2.156	no
OL+BDC	0.5	0.0869	2.300	yes
OL+BDC	1.0	1.4385	2.344	yes

Three observations follow.

First, $c_{bm} = 0.35$ is **not** a safe ceiling for `orig`. Σ is non-PD in five of the nine design-by-half-life cells: all three CO cells, and the zero-carryover cells of both other designs. *(Corrected 2026-10-07 from "four of the nine", which disagreed with the table and with Section 1.)* The safe ceiling across the full grid is 0.30, not 0.35. The exact ceilings are 0.346 in every zero-carryover cell and in CO, and 0.351 (Hybrid and OL+BDC) at $t_{1/2} = 0.5$ (verified by closed form, Section 5.1 note).

Second, where `orig` is PD at 0.35, it is marginal. Minimum eigenvalues of 0.083 and 0.087 at $t_{1/2} = 0.5$ sit 26.5- to 27.6-fold below `covar`'s at the same cells (an earlier version said three orders of magnitude; corrected 2026-10-07; "about 28-fold" corrected 2026-10-08 to the ratios of the table, 2.300/0.0869 = 26.5 in OL+BDC and 2.297/0.0831 = 27.6 in Hybrid), and well below `orig`'s own values at $c_{bm} = 0.10$. These are eigenvalues of the covariance matrix, so their size also reflects the component variances. A margin that thin is not robust to changes in ρ, `c.cfct`, the number of occasions, or the response parameters, none of which were varied in this sweep.

Third, the CO design is uniformly the binding case under `orig`, and its minimum eigenvalue does not vary with $t_{1/2}$ at all. The more likely reason is that CO's binding path is the placebo-first path, whose occasions before first exposure have zero coupling at every half-life and which never discontinues, so its coupling pattern does not change with carryover (inferred; the same mechanism is verified for the step coupling in `docs/34`, Section 4.6). An earlier version attributed this to the fast `orig` decay driving the off-drug entries of the drug-first path to zero.

4.3 Silent repair

Status: verified.

Both implementations call `make.positive.definite(sigma, tol = 1e-3)` when the PD check fails, with no error and no record in the returned object, and with no warning unless (in the package only) `verbose = TRUE` (`R/generateData.R` lines 376 to 389, inspected 2026-10-07):

```
if (makePositiveDefinite) {
  if (!is.positive.definite(sigma)) {
    sigma <- make.positive.definite(sigma, tol = 1e-3)
  }
}
```

The consequence is that a non-PD cell does not fail. It silently becomes a different DGP. Measuring the realized biomarker-response correlations after repair, for CO path A at $t_{1/2} = 1.0$:

Table 9: Minimum eigenvalue and realized correlation after silent repair by nominal c_{bm}, CO path A

nominal c_{bm}	min eigenvalue	realized max	repaired
0.20	5.5645	0.2000	no
0.25	5.0361	0.2500	no
0.30	3.6264	0.3000	no
0.35	0.8645	0.3500	no
0.40	-2.8964	0.3955	yes
0.45	-7.3385	0.4370	yes

The distortion is modest in magnitude, 0.005 and 0.013 in correlation in this example, but it is undisclosed and it is not constant across the grid, so it introduces a systematic and unreported difference between nominal and realized effect size precisely in the cells where `orig` is under the most strain. Power reported against a nominal c_{bm} in a repaired cell is not power at that c_{bm}.

4.4 Answer

Restricting c_{bm} to 0.35 or below does not remove the case for `covar`. Five reasons remain, in descending order of force.

1. **The restriction is not tight enough.** At exactly 0.35, `orig` is non-PD in five of nine cells, including every CO cell (corrected from "four",

2026-10-07). At the published $\rho = 0.8$ the restriction would have to be tighter still: the published construct's ceiling is 0.256 without carryover (`docs/36`). A restriction that actually cleared the PD objection would be $c_{bm} \leq 0.30$, and even then only if no other parameter moves.

2. **The `scalefactor` default is c_{bm}-independent and wrong.** `orig` decays the response mean at twice the nominal rate at every correlation strength (Section 3.5), understating residual exposure by a factor of two at the first off-drug occasion and by a factor of 16 at the off-drug occasion four weeks after COd (corrected from "up to 38", Section 3.5). The two processes are therefore not matched on carryover at a matched nominal $t_{1/2}$, so any comparison between them confounds carryover severity with everything else.

3. **The dropped first occasion is c_{bm}-independent.** It removes an on-drug occasion from the interaction channel at every parameter setting, including the null.

4. **Silent repair breaks the correspondence between nominal and realized effect size** in exactly the region where a restricted study would be operating closest to the boundary.

5. **The fidelity argument does not survive inspection, but not in the way it first appears.** The strongest reason to prefer `orig` at any c_{bm} is that it reproduces the published reference. The vendored file does not: it is upstream HEAD (`06dac83`, 2024-11-27), four years after the paper, not the pre-acceptance commit the directory README names. So `orig` as run is not Hendrickson et al. as published.

   The remedy is not to restore the published version. Section 3.3.1 shows the 2020 interaction gate is defective: it raises the off-drug biomarker-response correlation to full strength as carryover grows, inverting the effect residual exposure should have. Pinning to the pre-acceptance commit would buy documentary fidelity at the cost of reinstating a scientifically wrong DGP.

   Note added 2026-10-07. The project has since resolved this differently: `orig` is now `58b32a9` itself, run as published, and the repair of the coupling rule is made as an explicit, documented code change to it (graded coupling, paper 01 Appendix B.1), so fidelity and correctness are reported separately rather than traded.

Section 5 qualifies the first of these five reasons. A crossed decomposition shows the PD objection is narrower than it appears: it is an objection to compound symmetry alone, and one change fixes it. The other four reasons are not PD arguments and are unaffected.

The fifth point deserves the most emphasis, because it dissolves the trade-off rather than resolving it. One would ordinarily accept a narrower feasible parameter range as the price of matching a published method. Here neither branch is

attractive: the current arm does not match the published method, and the published method is not worth matching. The way out is a corrected process rather than a choice between the two existing ones, which is the subject of Section 4.5.

4.5 What would change the answer

Rather than choosing between the two existing processes, the better course is a corrected `orig`. Assessing the five 2024 changes individually (Sections 3.3 to 3.5) shows they are not of a piece: three are sound and two are defective.

Keep. The `brtest` / `rawbrmeans` capture (Difference 5's infrastructure), which correctly separates pre-carryover state from post-carryover mean. The interaction gate replacement, which repairs the inversion in the published code (Section 3.3.1). The `verbose` diagnostic blocks, which are inert with respect to returned values and are the mechanism by which the `scalefactor` problem would have been caught.

Fix. Set `scalefactor = 1`, or remove the parameter (Section 3.5). Remove the `p > 1` guard so occasion 1 is assigned (Section 3.4), which requires handling `p = 1` explicitly since the ratio has no predecessor there.

Fix separately. The carryover recursion double-counts elapsed time by applying $(1/2)^{t_{sd}/t_{1/2}}$, with cumulative t_{sd}, to an already-adjusted value. This is inherited from the published code and is present in `covar` as well, so correcting it changes both processes. *(Since done for `covar` and the mean-moderation process; Section 3.5, note.)*

A process so corrected would be better than the published original, better than current HEAD, and would isolate the correlation structure as the substantive remaining difference from `covar`. Combined with the single AR(1) change of Section 5, it would also resolve the PD objection outright.

Two further conditions would independently strengthen any comparison:

- Restricting the study to $c_{bm} \leq 0.30$ and to designs with $t_{1/2} > 0$, under which the PD objection does not arise even without the AR(1) change. (This holds at $\rho = 0.7$. At the published $\rho = 0.8$ it does not: the CO ceiling is 0.256 at every half-life, `docs/36`, Section 4.2.)
- Running the correlation structure and the interaction gate as crossed factors, as Paper 01's own Appendix B.6 suggests. Section 5 does this for positive definiteness; the power half remains undone.

5. Decomposition: the minimum set of changes that fixes PD

5.1 Motivation and method

The stated reason for the `covar` changes was the non-positive-definite correlation matrix under `orig`. That justification implies a counterfactual which has not previously been tested: how many of the four changes are actually required to fix PD, and does each contribute in the direction assumed?

Status: verified. A single Σ constructor was written with one flag per candidate change, reproducing `orig` when all four flags are off and `covar` when all four are on. The flags are:

- `ar1`: AR(1) within-factor, $\rho^{|w_t - w_s|}$, versus compound symmetry (constant ρ). Difference 1.
- `cfd`: decaying cross-factor off-diagonal, $c_\times \rho^{|w_t - w_s|}$ with $c_\times =$ `c.cfct`, versus constant $c_\times$. Difference 2.
- `bexp`: exponential coupling, $c_{bm} e^{-\lambda t_{sd}}$, versus the ratio-graded `orig` form. Difference 3.
- `occ1`: assign the biomarker-response correlation at occasion 1 versus skipping it. Difference 4.

All 16 combinations were crossed with three designs and three half-lives. For each, c_{bm} was swept from 0.05 to 0.80 in steps of 0.01, and the largest value at which Σ remained PD across every randomization path was recorded. The reported ceiling for a configuration is its worst cell over all nine design-by-half-life combinations.

5.2 Effect of each change in isolation

Starting from `orig` (all four flags off, worst-cell ceiling 0.34) and turning on exactly one flag:

Table 10: Worst-cell ceiling with exactly one change enabled, starting from orig

change enabled	worst-cell ceiling	effect
none (`orig`)	0.34	baseline
`ar1` only	**0.48**	**+0.14**
`cfd` only	0.16	-0.18
`bexp` only	0.34	none
`occ1` only	0.34	none

The result is unambiguous. **The AR(1) change alone accounts for the entire PD improvement.** The other three contribute nothing positive, and one of them is actively harmful.

5.3 The minimum set is a single change

`ar1` is both necessary and sufficient. Every configuration with `ar1 = TRUE` reaches a worst-cell ceiling of 0.43 or above; every configuration with `ar1 = FALSE` is capped at 0.34 or below, regardless of what the other three flags are set to. Ranked across all 16 configurations, the eight `ar1 = TRUE` configurations occupy the top eight positions without exception.

Per design and half-life:

Table 11: Ceiling by design and half-life for orig, AR(1) only and full covar

design	$t_{1/2}$	orig	AR(1) only	full covar
CO	0.0	0.34	0.59	0.61
CO	0.5	0.34	0.59	0.61
CO	1.0	0.34	0.59	0.61
Hybrid	0.0	0.34	0.49	0.45
Hybrid	0.5	0.35	0.51	0.50
Hybrid	1.0	0.36	0.54	0.54
OL+BDC	0.0	0.34	0.48	0.45
OL+BDC	0.5	0.35	0.49	0.49
OL+BDC	1.0	0.36	0.52	0.53

Changing only the within-factor correlation form, and leaving the other three `orig` behaviors untouched, yields a higher worst-cell ceiling (0.48) than the full set of four changes (0.45). The minimal fix is not merely sufficient; it is strictly better than `covar` on the criterion that motivated `covar` in the first place. AR(1)-only matches or exceeds full `covar` in seven of the nine cells, and trails it by at most 0.01 in the remaining two.

Correction (2026-10-07). The advantage of "AR(1) only" over the full set is an artifact of Difference 4. "AR(1) only" keeps `occ1` off, so its first occasion, which is on drug in every Hybrid and OL+BDC path, carries no coupling; fewer coupled occasions give a smaller $v^\top M^{-1} v$ and a higher ceiling. With the first occasion coupled, AR(1) with a constant cross-factor block has a no-carryover ceiling of 0.452 in Hybrid and 0.438 in OL+BDC, below the full set's 0.456 and 0.452 (worst cell 0.438 against 0.452). Without it, the same computation reproduces this table's AR(1)-only column: 0.497 and 0.482 (verified by an independent closed-form computation, 2026-10-07; the 2024 ratio-graded coupling and the step coupling coincide at $t_{1/2} = 0$). The conclusion that AR(1) is the one change that matters for positive definiteness stands; the claim that it is "strictly better than `covar`" does not, since it is bought by leaving an on-drug occasion uncoupled.

(Clarified 2026-10-08.) "AR(1) only" here means AR(1) within components with the published constant cross-factor block $c_\times$ (`cfd` off). `docs/36`

and `docs/44` use "AR(1) alone" for their configuration B, which applies AR(1) to the cross-factor block as well, $c_{\times}\rho^{|w_t - w_s|}$. The two are different constructs: at $\rho = 0.7$, no carryover and the first occasion coupled, they give 0.438 (constant block) and 0.452 (decaying block) in OL+BDC. See `analysis/report/compendium-mathematics.tex`, Section "The 2024 revision of the published code: which change fixes the ceiling".

5.4 Why the cross-factor change is harmful

The `cfd` change, applied without `ar1`, lowers the ceiling from 0.34 to 0.16, the worst configuration tested. The mechanism is structural incoherence. Under compound symmetry a factor's correlation with itself is a constant ρ at every lag. Introducing a decaying cross-factor term makes a factor's correlation with a *different* factor fall away with lag while its correlation with *itself* does not. At long lags the matrix asserts strong within-factor coupling alongside near-zero cross-factor coupling, and the resulting block structure drives the smallest eigenvalue negative sooner than either uniform choice does.

Under `ar1` the same change is close to neutral, contributing at most 0.02 in either direction, because within-factor and cross-factor terms then decay together and the matrix stays internally consistent.

This is worth stating plainly: **Difference 2 was not a PD fix. In isolation it is a PD regression, and its only defensible justification is internal consistency with Difference 1.**

5.5 Why the coupling and occasion-1 changes are irrelevant to PD

Neither `bexp` nor `occ1` moved the ceiling by a single grid step in any configuration. This is expected on inspection. The biomarker-response entries occupy one row and one column of a matrix with $2 + 3n$ dimensions, and their magnitudes are bounded by c_{bm} itself. The binding constraint comes from the $3n \times 3n$ within- and cross-factor blocks, not from the biomarker margin.

Correction (2026-10-07). Both statements in the preceding paragraph are wrong. First, the changes do move the ceiling once `ar1` is on: `occ1` lowers the no-carryover ceiling from 0.497 to 0.452 (Hybrid) and from 0.482 to 0.438 (OL+BDC), and at positive half-lives the coupling form moves it by several hundredths (verified, Section 5.3 note). They leave the *worst cell* unchanged only when `ar1` is off, where it is 0.346 either way (under compound symmetry the ceiling of a 0/1 coupling pattern depends only on how many occasions are coupled, `docs/34`, Section 5.1). Second, the mechanism is the reverse of the one stated. The response block is positive definite in every configuration examined (smallest eigenvalues 0.015 to 0.33, `docs/34`, Table 4.2), and by the Schur complement Σ fails only when $c_{bm}^2 v^\top M^{-1} v \geq 1$, so the biomarker row is always the binding constraint, and the ceiling depends on the coupling pattern

v as much as on the correlation structure (`docs/34`, Section 3.2; `docs/36`, Section 4.2).

The practical consequence drawn below survives in substance. Dropping occasion 1 raises the ceiling under AR(1) only by removing the coupling from an on-drug occasion, which is not a defensible way to gain range, and Difference 3 does not move the worst cell. In that sense Differences 3 and 4 cannot be defended on PD grounds at all. Difference 3 is an undocumented change of decay rate (Section 3.3) and Difference 4 is a defect (Section 3.4). Whatever their merits, the non-positive-definiteness of the `orig` matrix is not among them.

5.6 Implications

1. **The minimum set of changes that fixes the PD problem is one change: compound symmetry to AR(1).** Adopting it alone recovers more feasible parameter range than the full four-change set. *(Qualified 2026-10-07: only while occasion 1 stays uncoupled; with it coupled, AR(1) alone reaches 0.438 against the full set's 0.452, Section 5.3 note. A separable AR(1) cross-factor block, `docs/35`, raises the ceiling further, to 0.474 at $\rho = 0.7$.)*
2. **Three of the four changes in `covar` are not PD fixes.** They may be justified on other grounds, but the PD rationale does not extend to them and should not be offered for them.
3. **A minimal-change variant is worth constructing.** An `orig + AR(1)` process would isolate the correlation structure as the sole difference from the published reference, which is exactly the crossed-factor decomposition Paper 01's Appendix B.6 says would be required, and it is a one-line change to the vendored file.
4. **This changes the practical answer of Section 4.4 in one respect.** The PD objection to `orig` is narrower than it appeared: it is an objection to compound symmetry specifically, not to the `orig` process as a whole. The other four objections in Section 4.4 stand unchanged, since none of them is a PD argument.

6. Recommendations

1. **Do not adopt `orig` as a comparison arm on the strength of a restricted c_{bm} range.** The restriction addresses one of four differences and does not fully address even that one.
2. **Correct Paper 01, Appendix B.5 to B.7.** The $\tilde{r}$ row, the worked example, and the third interpretive paragraph after the B.6 table describe code that does not run. This is independent of any decision about future simulation work and should be treated as a correctness fix.

3. **Correct the directory README's provenance paragraph.** The vendored base is upstream HEAD `06dac83` (2024-11-27), not `3035581`, and the date given for that commit (2026-06-28) is wrong in year (the commit is 2020-06-28 04:00 UTC, which is 2020-06-27 in US Pacific time; an earlier version said the day was also wrong, which depends on the time zone). The 'no other lines were changed' claim is accurate once the correct base is named. The `RON THOMAS` annotations are upstream, from `8609f12`, and should not be described as local modifications.

3b. **Do not revert to the published version to resolve this.** Section 3.3.1 shows the 2020 interaction gate is defective. Pinning to `3035581` would buy documentary fidelity at the cost of a scientifically wrong DGP. Correct the description and the two remaining implementation flaws instead (Section 4.5).

3c. **Report the `Dbc` never-treated bug upstream** (Section 3.6.2). It is a genuine correctness defect in `rchendrickson/pmsimstats` HEAD, it is independent of everything else in this document, and the fix already exists in this package's `R/lme_analysis.R` and can be offered directly. Until it is fixed, the `orig` arm must not be run with `carryover_t1half > 0`, since its crossover results would be meaningless.

4. **Make PD repair loud.** Both implementations should record the repair in the returned object and warn. A silent change to the DGP is not acceptable in a simulation study whose outputs are reported against nominal parameter values.

5. **Fix the `p > 1` guard** if `orig` is retained in any form, or document it as a known property of the arm if the intent is to reproduce whatever the vendored file does.

6. **Construct an `orig + AR(1)` variant.** Section 5 shows this is a one-line change that fixes PD more effectively than the full `covar` set (but see the Section 5.3 note: the margin over `covar` comes from the uncoupled first occasion). It isolates the correlation structure as the sole difference from the reference implementation and would let Paper 01 report a decomposition rather than a confounded bound.

7. **Stop offering the PD rationale for Differences 2, 3, and 4.** Section 5 shows none of them improves positive definiteness, and Difference 2 degrades it when applied on its own. If those changes are retained they need a different justification.

8. **If a full decomposition of the power results is wanted, run the crossed design.** Correlation structure (CS versus AR(1)) crossed with interaction gate (step versus graded) at $c_{bm} \leq 0.30$, three designs, three half-lives. This is four DGP variants rather than two and would separate the effects the present comparison confounds. The PD half of that decomposition is already done (Section 5); the power half is not.

7. Limitations

The PD sweep used a single parameter set ($\rho = 0.7$, `c.cf1t = 0.2`, `c.cfct = 0.1`, `N = 35`, eight occasions, the package's `extracted_rp` and `extracted_bp`). The PD boundary depends on all of these, and the boundaries reported in Section 4.2 should not be read as general. In particular, larger ρ will tighten the `orig` boundary further. *(Confirmed 2026-10-07: at the published $\rho = 0.8$ the compound-symmetry ceiling is 0.256 without carryover, `docs/36`, Section 4.2.)*

The Section 5 decomposition is a statement about positive definiteness only. It establishes which changes widen the feasible c_{bm} range, not which produce better statistical behavior within that range. An `orig + AR(1)` process has a higher PD ceiling than `covar` (while its first occasion stays uncoupled; Section 5.3 note) but was not evaluated for power, Type I error, or estimator bias, and nothing here implies it would perform better on those.

The Section 5 constructor reproduces both processes from the source read in Section 2 rather than calling them. It was checked against the `orig` and `covar` ceilings obtained in Section 4.2 from the real implementations, which agree to within the coarser grid step used there, but it is a reimplementation and may diverge in behavior not exercised by this sweep.

No power simulation was run for this document. The consequences of Differences 3 and 4 for realized power under `orig` are argued from the structure of the covariance matrix, not measured. Quantifying them would require running the `orig` arm with and without the `p > 1` guard, and with the commented-out gate restored, which is the natural follow-up.

The upstream provenance question is now settled by direct comparison (Section 2), but one part of it is not. Commit `8609f12` is authored and committed by RC Hendrickson under the message 'Ron Thomas' edits, currated into working code'. The history cannot show which parts of that commit originated with Ron Thomas and which arose in the curation, so responsibility for the `scalefactor` default and the `p > 1` guard cannot be assigned from the record. This matters only for deciding with whom to raise them.

The upstream code has not been executable on a current stack for some time: the pre-2021 `data.table` indexing idiom is present in all 21 commits including HEAD, so `generateData()` fails outright without the vendored patch. The `orig` arm therefore reproduces upstream source but not upstream behavior, since upstream does not run unmodified.

`implementations/original/R/generateData.R`, the package's other copy of the reference implementation, was inspected and found to differ from the vendored copy: it gates on `d[p]$onDrug` and uses `c.cfct * rho^tg` for the cross-factor off-diagonal. It is therefore a third variant, distinct from both `covar` and the vendored `orig`, and its relationship to the other two was not pursued here.

8. Provenance

Audit conducted 2026-09-08 against the working tree at commit `3ce1162`. Computations executed on the host R session (R 4.6.1), outside the project container. Scratch scripts were not retained in the repository.

The 2026-10-07 review corrections were checked with an independent base-R closed-form ceiling computation (not retained in the repository) and with `gh api` queries of the upstream commit history.

References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020;2:13. doi:10.3389/fdgth.2020.00013. (Verified via Crossref, 2026-10-07.) Code: `github.com/rchendrickson/pmsimstats`; the vendored copy discussed here is HEAD `06dac83`, and the published state is commit `58b32a9`.